

POINT DEFECTS IN CRYSTALS

- ❑ Overview
- ❑ Vacancies & their Clusters
- ❑ Interstitials
- ❑ Defects in Ionic Crystals
 - Frenkel defect
 - Shottky defect

Part of

MATERIALS SCIENCE
& *A Learner's Guide*
ENGINEERING

AN INTRODUCTORY E-BOOK

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<http://home.iitk.ac.in/~anandh/E-book.htm>

Advanced Reading

Point Defects in Materials

F. Agullo-Lopez, C.R.A. Catlow, P.D. Townsend

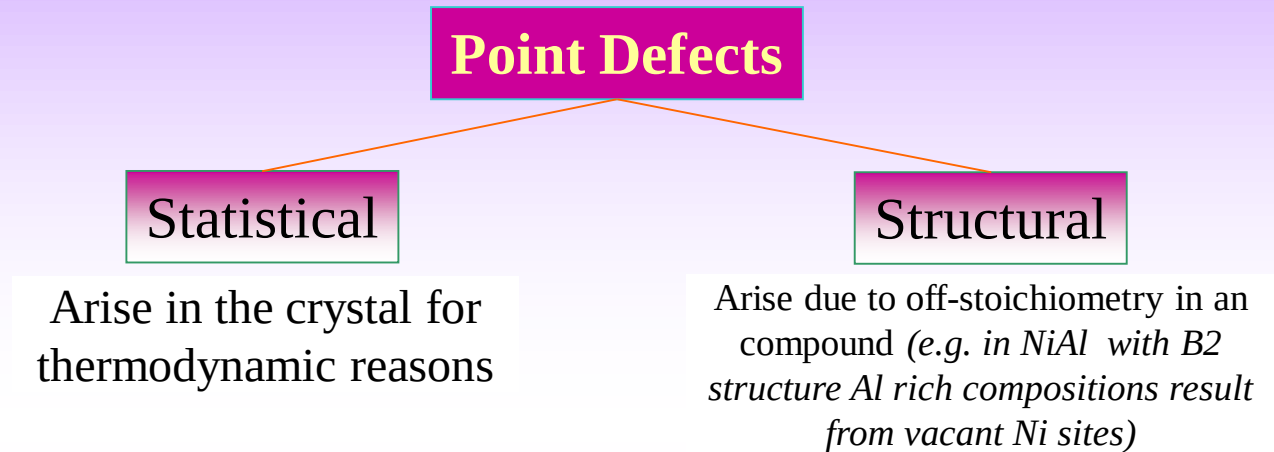
Academic Press, London (1988)

Caution Note: In any chapter, amongst the first few pages (say 5 pages) there will be some 'big picture' overview information. This may lead to 'overloading' and readers who find this 'uncomfortable' may skip particular slides in the first reading and come back to them later.

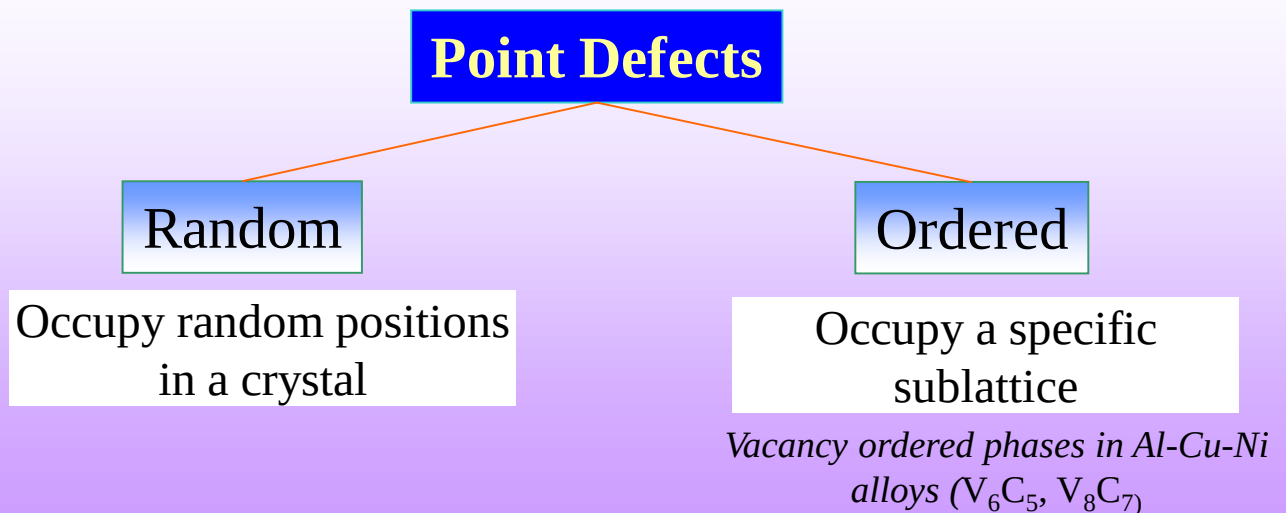
- ❑ Point defects can be considered as 0D (zero dimensional) defects.
- ❑ The more appropriate term would be '*point like*' as the influence of 0D defects spreads into a small region around the defect.
- ❑ Point defects *could* be associated with stress fields and charge.
- ❑ Point defects could associate to form larger groups/complexes
→ the behaviour of these groups could be very different from an isolated point defect.
➤ *E.g. in the case of vacancy clusters in a crystal plane the defect could be visualized as an edge dislocation loop.*
- ❑ Point defects could be associated with other defects (like dislocations, grain boundaries etc.)
➤ Segregation of Carbon to the dislocation core region gives rise to yield point phenomenon.
➤ 'Impurity'/solute atoms may segregate to the grain boundaries.
- ❑ Based on **Origin** Point defects could be **Random** (statistically stored) or **Structural**.
More in the next slide.
- ❑ Based on **Position** Point defects could be **Random** (based on position) or **Ordered**.
More in the next slide.
- ❑ Point defects may play a profound role in many of the material properties like the following.
➤ Vacancies 'carry' the diffusion of substitutional atoms.
➤ F-centres (*Farbzentrum*) determine the colour of many crystals (e.g. in LiCl and KBr).
➤ The presence of interstitial C atoms increases the hardness of martensite.

- ❑ Point defects can be classified as below from two points of view.
- ❑ The behaviour of a point defect depends on the class (as below) to which it belongs.

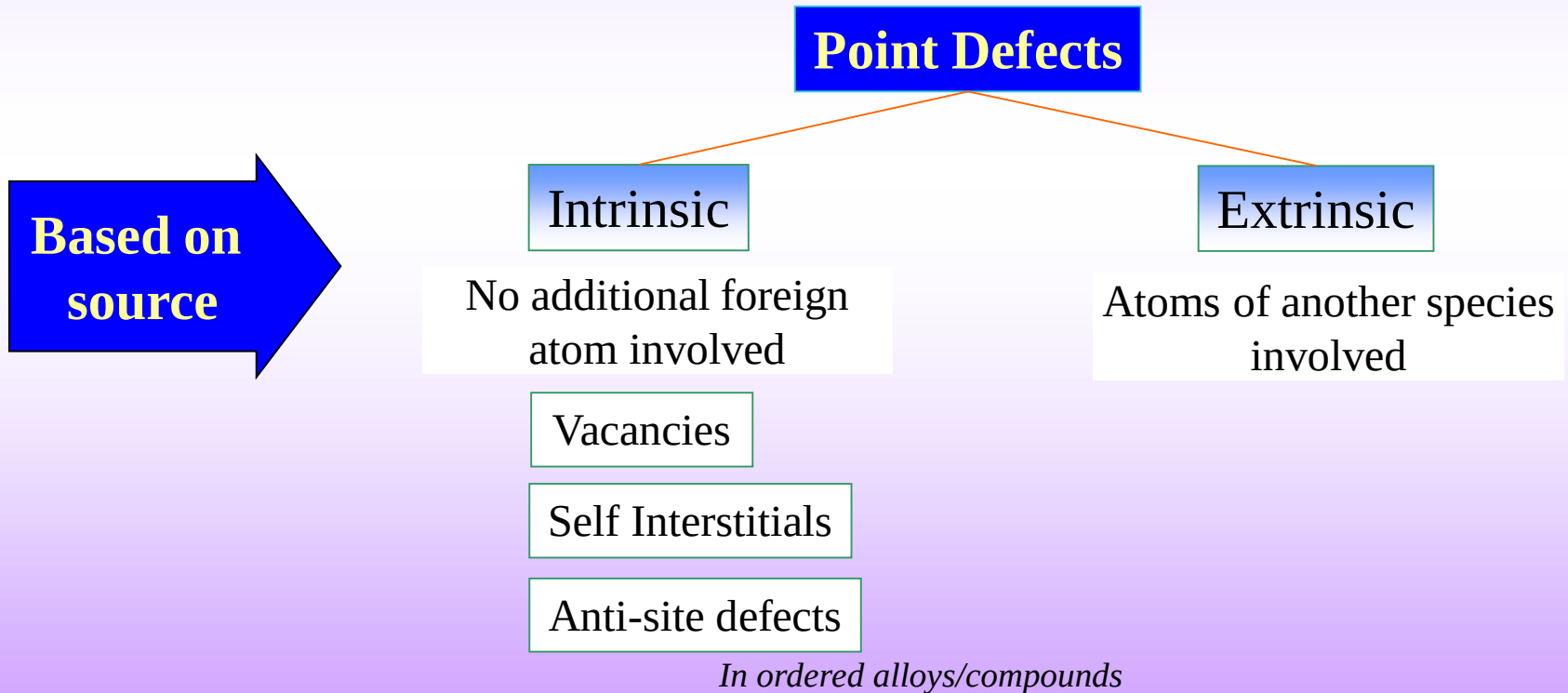
**Based on
origin**



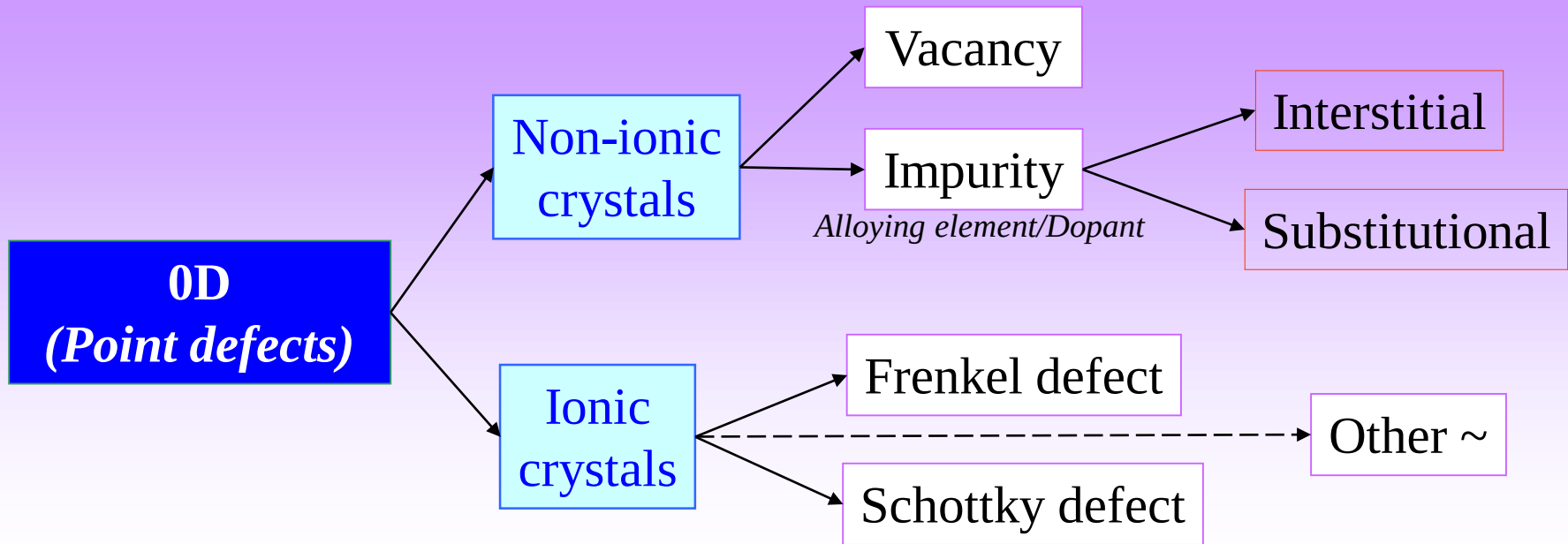
**Based on
position**



❑ This classification based on source is simple to understand.



Note: Presence of a different isotope may also be considered as a defect



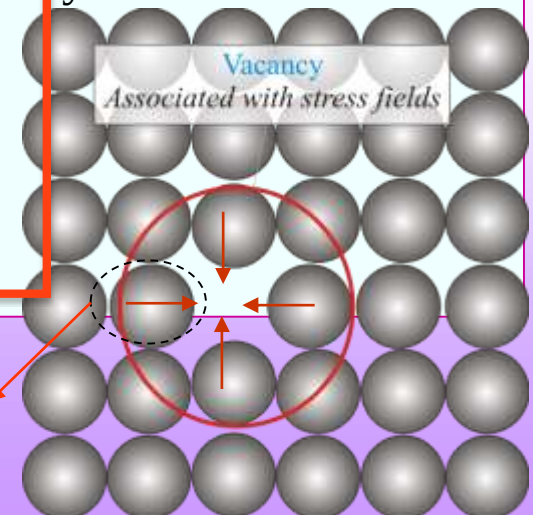
- ❑ 0D (point) defects are imperfect *point-like* regions in the crystal about the size of 1-2 atomic diameters.
- ❑ The extent of the distortion field may however extend to a larger distance.
- ❑ Point defects can be created by ‘removal’, ‘addition’ or displacement of an atomic species (atom, ion).
- ❑ Defect structures in ionic crystals can be more complex and are not discussed in detail in the elementary introduction.

Vacancy

- ❑ Missing atom from an *atomic site** is called a vacancy.
- ❑ Atoms around the vacancy displaced from their equilibrium positions.
- ❑ This gives rise to a stress field in the vicinity of the vacancy.
- ❑ Based on their origin vacancies can be:
 - **Random/Statistical** (**thermal vacancies**, which are required by thermodynamic equilibrium) or
 - **Structural** (due to off-stoichiometry in a compound).
- ❑ Based on their position vacancies can be **random** or **ordered**. (*Ordered defects become part of the crystal structure and are 'no longer defects' in the usual sense*).
- ❑ Vacancies play an important role in diffusion of substitutional atoms and in many other processes/effects in materials science, including climb of edge dislocations, some forms of creep and increased resistivity.
- ❑ Non-equilibrium concentration of vacancies can be generated by:
 - quenching from a higher temperature
 - bombardment with high energy particles
 - plastic deformation.
 - off-stoichiometry in ordered compounds.*Etc.*

Random

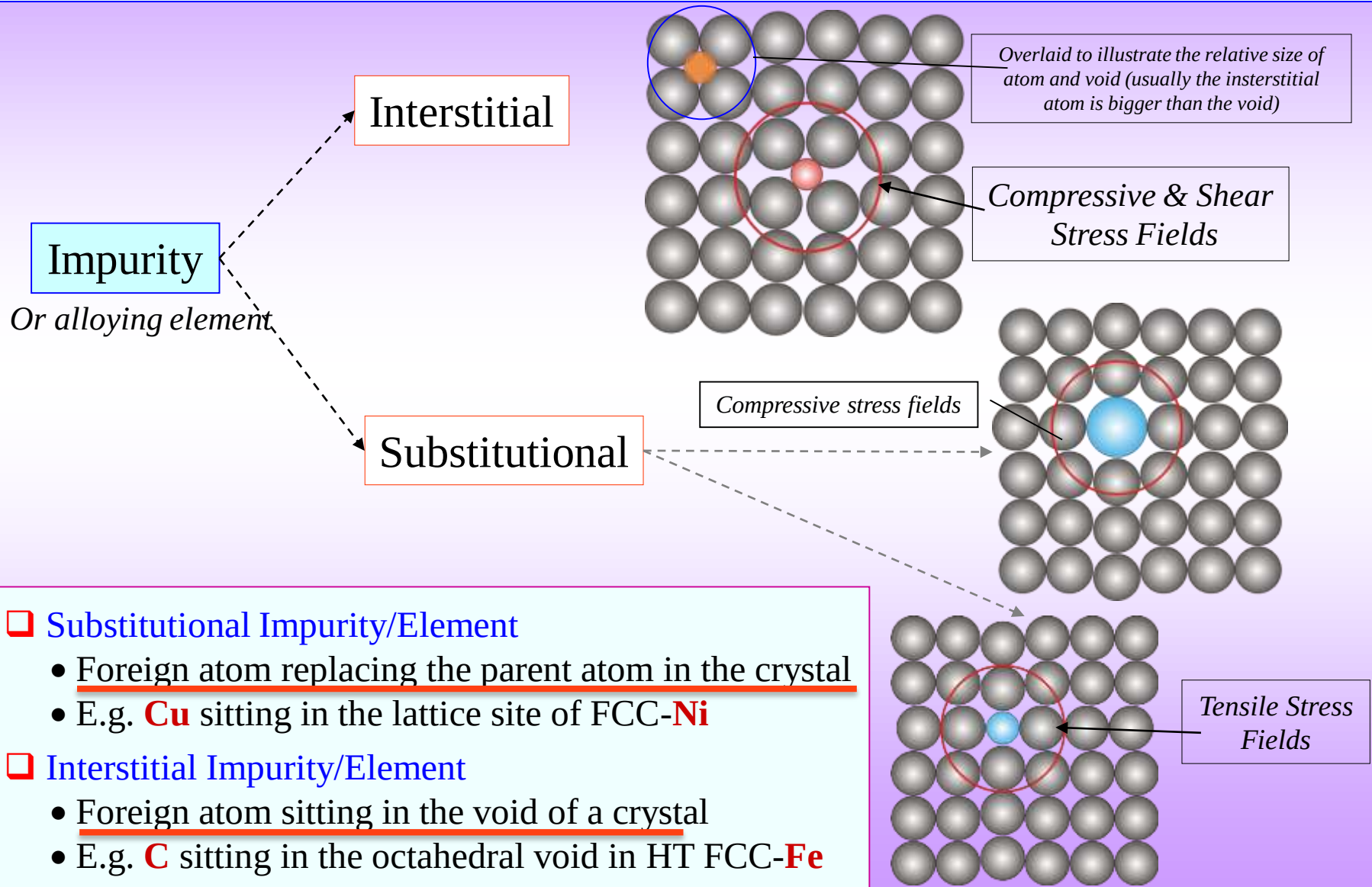
Structural



Neighbouring atoms are displaced from their equilibrium position in a perfect crystal

* *Note: we have not used the term "lattice site".*

- ❑ A 'foreign' element added (called as impurity/alloying element/dopant based on the context) can go to an interstitial site (between atoms) or may substitute for an atom of the host.



- ❑ In some (rare) situations the same element can occupy both a lattice position and an interstitial position ► e.g. B in steel.
- ❑ By using ion irradiation or some other ‘strong forces’ substitutional atoms may be forced to occupy an interstitial position.
- ❑ The diffusion mechanism of these two types of point defects (interstitial vs substitutional) is different. This is because for the diffusion of substitutional atom the neighbouring site has to be vacant; while in the case of interstitial diffusion the neighbouring site is usually vacant (as the solubility of interstitial atoms is small).

Interstitial C sitting in the octahedral void in HT FCC-Fe

- ❑ $r_{\text{Octahedral void}} / r_{\text{FCC atom}} = 0.414$
- ❑ $r_{\text{Fe-FCC}} = 1.29 \text{ \AA} \Rightarrow r_{\text{Octahedral void}} = 0.414 \times 1.29 = 0.53 \text{ \AA}$
- ❑ $r_{\text{C}} = 0.71 \text{ \AA}$
- ❑ $\Rightarrow$ **Compressive strains** around the C atom
- ❑ Solubility limited to **2 wt%** (9.3 at%)

Interstitial C sitting in the octahedral void in LT BCC-Fe

- ❑ $r_{\text{Tetrahedral void}} / r_{\text{BCC atom}} = 0.29 \bullet r_{\text{C}} = 0.71 \text{ \AA}$
- ❑ $r_{\text{Fe-BCC}} = 1.258 \text{ \AA} \Rightarrow r_{\text{Tetrahedral void}} = 0.29 \times 1.258 = 0.364 \text{ \AA}$
- ❑ **But C sits in smaller octahedral void- displaces fewer atoms**
- ❑ $\Rightarrow$ **Severe compressive strains** around the C atom
- ❑ Solubility limited to **0.008 wt%** (0.037 at%)

Why are vacancies referred to as equilibrium thermodynamic defects?

In these discussions we will keep in view metallic crystals like Fe, Cu, Zn, etc.

Here equilibrium implies stable state (with lowest G)

- ❑ Formation of a vacancy leads to 'missing bonds' and distortion of the lattice.
- ❑ Hence, the potential energy (Internal energy & Enthalpy) of the system increases.
- ❑ Work required for the formation of a point defect →
Enthalpy of formation $\{\Delta H_f = (H_{\text{crystal with one vacancy}} - H_{\text{perfect crystal}})\}$ [kJ/mol or eV/defect].
- ❑ Though it costs energy to form a vacancy, its formation leads to increase in **configurational entropy** (the crystal without vacancies represents just one state, while the crystal with vacancies can exist in many energetically equivalent states, corresponding to various positions of the vacancies in the crystal → 'the system becomes configurationally rich').
- ❑ ⇒ at some temperature above zero Kelvin there is an **equilibrium** concentration/number of **vacancies** (at 'low' temperatures no vacancies may be stable). *Refer to a calculation later for the calculation of the T at which the first vacancy becomes stable.*
- ❑ These type of vacancies are called **Thermal Vacancies** (and will not leave the crystal on annealing at a temperature at which these are stable → Thermodynamically stable).

- Note: up and above the equilibrium concentration of vacancies, there might be a additional non-equilibrium concentration of vacancies which are present. This can arise by quenching from a high temperature, irradiation with ions, cold work etc.
- When we quench a sample from high temperature *part of the* higher concentration of vacancies present (at higher temperature there is a higher equilibrium concentration of vacancies present), may be quenched-in at low temperature.

Enthalpy of formation of vacancies (ΔH_f)

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

Crystal	Kr	Cd	Pb	Zn	Mg	Al	Ag	Cu	Ni
kJ / mol	7.7	38	48	49	56	68	106	120	168
eV / vacancy	0.08	0.39	0.5	0.51	0.58	0.70	1.1	1.24	1.74

Note that the second row is in kJ per mole of vacancies while the 3rd row is eV per vacancy.

Calculation of equilibrium concentration of vacancies

Revise chapter on equilibrium before this computation

- ❑ Let n_v be the number of vacancies, N the number of sites in the lattice
- ❑ Assume that concentration of vacancies is small i.e. $n_v/N \ll 1$
 - $\Rightarrow$ the interaction between vacancies can be ignored ($\Rightarrow \frac{\partial \Delta H_f}{\partial n_v} = 0$) (α)
 - $\Rightarrow \Delta H_{\text{formation}}(n_v \text{ vacancies}) = n_v \cdot \Delta H_{\text{formation}}(1 \text{ vacancy})$
- ❑ Let ΔH_f be the enthalpy of formation of one vacancy (assumed constant for now).

$$\Delta G = (G_{\text{crystal with vacancies}} - G_{\text{perfect crystal}}) = \Delta H - T \Delta S$$

$$(1) \Delta S = \Delta S_{\text{configurational}} = (S_{\text{state with vacancies}} - S_{\text{state without vacancies=perfect crystal}})$$

$$\Delta G (\text{putting } n \text{ vacancies}) = n_v \cdot \Delta H_f - T \Delta S_{\text{config}} \leftarrow \text{Configurational entropy}$$

❑ Calculating ΔS_{config} :

- $S = k \cdot \ln(\omega)$. For the state without vacancies (perfect crystal), the number of configurations is '1' $\Rightarrow S_{\text{perfect crystal}} = k \cdot \ln(1) = 0$. Hence, $\Delta S_{\text{configurational}} = S_{\text{state with vacancies}}$.
- In a lattice with N atoms (which could be $N_{\text{Avagadro}} = N_0$) there are n_v vacancies and $(N - n_v)$ filled sites. The possible number of configurations (ω) is given by: ${}^N C_{n_v}$
(i.e. the possible number of ways I can chose n_v vacant sites from a perfect lattice containing N sites).

$${}^N C_{n_v} = \frac{N!}{n_v! (N - n_v)!} = \omega$$

$$\Delta S_{\text{config}} = k \ln \omega = k \cdot \ln \left\{ \frac{N!}{n_v! (N - n_v)!} \right\} = k \left[\ln(N!) - \ln(n_v!) - \ln\{(N - n_v)!\} \right]$$

Continued...

(2)

From equations (1), (2)

$$\Delta G = n_v \cdot \Delta H_f - T \Delta S_{config} = \Delta H_f - kT \left[\ln(N!) - \ln(n_v!) - \ln\{(N - n_v)!\} \right]$$

Using Sterling's approximation:

$$\Delta G = n_v \cdot \Delta H_f - kT \left[\{N \ln(N) - N\} - \{n_v \ln(n_v) - n_v\} - \{(N - n_v) \ln(N - n_v) - (N - n_v)\} \right]$$

$$\ln[n] = [n \ln(n) - n]$$

$$\Delta G = n_v \cdot \Delta H_f - kT \left[N \ln(N) - n_v \ln(n_v) - (N - n_v) \ln(N - n_v) \right]$$

For energy minimum $\rightarrow \frac{\partial \Delta G}{\partial n_v} = 0$

$$\left. \frac{\partial \Delta G}{\partial n_v} \right|_T = \Delta H_f + n_v \frac{\partial \Delta H_f}{\partial n_v} - T \frac{\partial \Delta S_{config}}{\partial n_v} = 0$$

(Note: $\frac{\partial \Delta H_f}{\partial n_v}$ is crossed out with a red line and labeled "zero from (α)" with an arrow pointing to it.)

$$\left. \frac{\partial \Delta G}{\partial n_v} \right|_T = \Delta H_f + T \frac{\partial \Delta S_{config}}{\partial n_v} = \Delta H_f - kT \frac{\partial \left[N \ln(N) - n_v \ln(n_v) - (N - n_v) \ln(N - n_v) \right]}{\partial n_v} = 0$$

$$\left. \frac{\partial \Delta G}{\partial n_v} \right|_T = \Delta H_f - kT \left[0 - \{(-1) \ln(N - n_v)\} + \left\{ \frac{(N - n_v)}{(N - n_v)} (-1) \right\} - \{ \ln n_v + n_v \cdot \frac{1}{n_v} \} \right]$$

$$\frac{\partial \Delta S_{config}}{\partial n_v} = k \ln \left(\frac{N - n_v}{n_v} \right)$$

$$\left. \frac{\partial \Delta G}{\partial n_v} \right|_T = \Delta H_f - kT [\ln(N - n_v) - \ln n_v] = \Delta H_f - kT \ln \left(\frac{N - n_v}{n_v} \right) = 0$$

$$\Rightarrow \frac{\Delta H_f}{kT} = \ln \left(\frac{N - n_v}{n_v} \right)$$

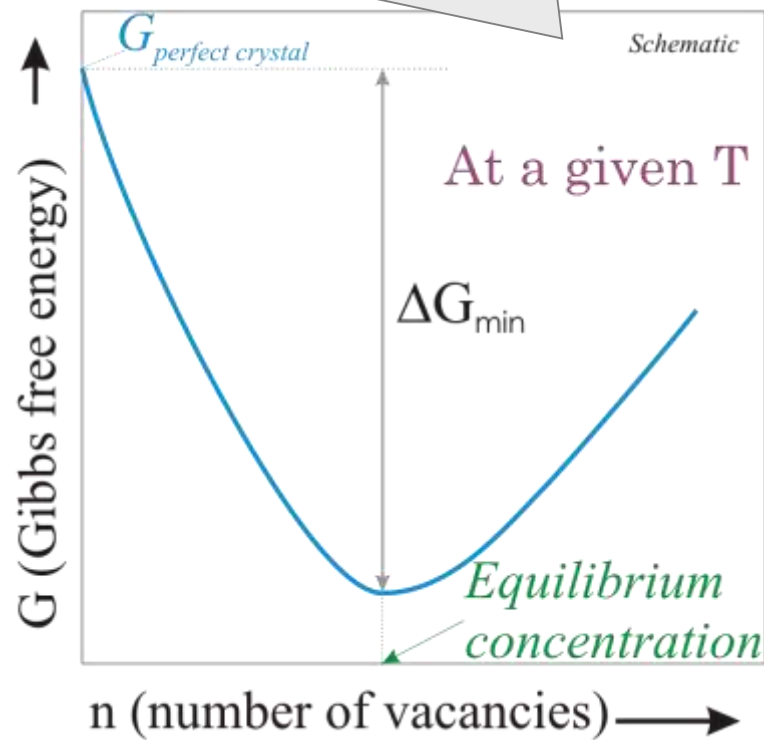
$$\text{Exp} \left(\frac{\Delta H_f}{kT} \right) = \frac{N}{n_v} - 1 \quad \text{Assuming } n_v \ll N$$

$$\frac{n_v}{N} = \exp \left(- \frac{\Delta H_f}{kT} \right)$$

$k = k_B = \text{Boltzmann constant}$
 $= 1.38 \times 10^{-23} \text{ J/K}$
 $= 8.62 \times 10^{-5} \text{ eV/K}$

User **R** instead of **k** if ΔH_f is in J/**mole** (instead of J/atom)

Variation of G with vacancy concentration at a fixed temperature



T (°C)	n/N
500	1×10^{-10}
1000	1×10^{-5}
1500	5×10^{-4}
2000	3×10^{-3}
$\Delta H_f = 1 \text{ eV/vacancy}$ $= 0.16 \times 10^{-18} \text{ J/vacancy}$	

- ❑ Close to the melting point in FCC metals Au, Ag, Cu the fraction of vacancies is about 10^{-4} (i.e. one in 10,000 lattice sites are vacant).

Metal	n/N at T_m
Cu	2×10^{-4}
Kr	3×10^{-3}
Cd	6.2×10^{-4}
Al	9×10^{-4}

- ❑ Even though it costs energy to put vacancies into a crystal (due to ‘broken bonds’), the Gibbs free energy can be lowered by accommodating some vacancies into the crystal due to the configurational entropy benefit that this provides.
- ❑ Hence, **certain equilibrium concentration/number of vacancies are preferred at $T > 0K$.**

Q & A

At what temperature does the first vacancy become stable in a Cu crystal?

- ❑ This we can determine by substituting $n_v = 1$ in the equation below we can determine the temperature.

$$\frac{n_v}{N} = \exp\left(-\frac{\Delta H_f}{kT}\right)$$

Data:

- No. of atoms in the crystal = N_{Avagadro}
- ΔH_f (Cu) = 1.24 eV/vacancy
- $k_B = 8.62 \times 10^{-5} \text{ eV/K}$
- $n_v = 1$

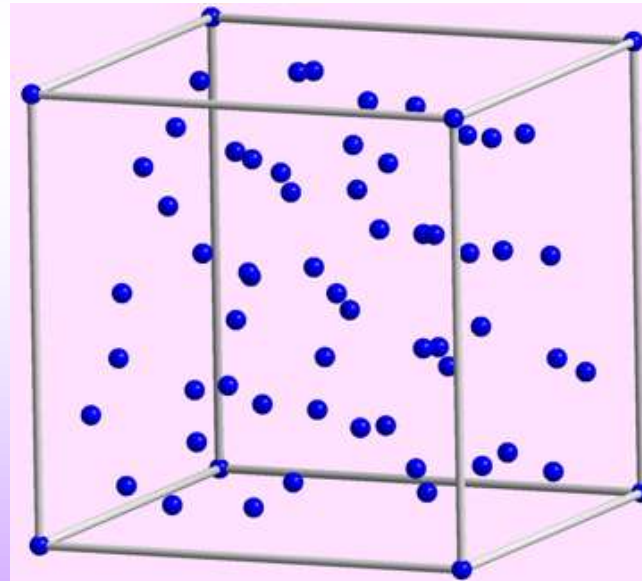
$$\ln\left(\frac{n_v}{N}\right) = -\frac{\Delta H_f}{kT}$$

$$T|_{\text{first vacancy}} = \frac{\Delta H_f}{k \ln N} = \frac{1.24}{(8.62 \times 10^{-5})(54.755)} = 262.72 \text{ K}$$

We assume that the missing atom goes to the surface. With the assumption that the number of surface sites is small– we need not worry about this one atom!

We had said: “Missing atom from an *atomic site* is called a vacancy”. Why did we say from an “atomic site” and did not say from a “*lattice site*”?

- ❑ In simple crystals with monoatomic decoration of lattices like SC, BCC or FCC, these two terms are equivalent (i.e. ‘atomic site’ $\equiv$ ‘lattice site’).
- ❑ In a crystal where many atoms are associated with the lattice, any of the missing atoms can be considered a vacancy. E.g. in α -Mn (BCC lattice, cI58) with 58 atoms in a unit cell, any of the missing atoms can be considered a vacancy.



α Mn

n = 58

cI58

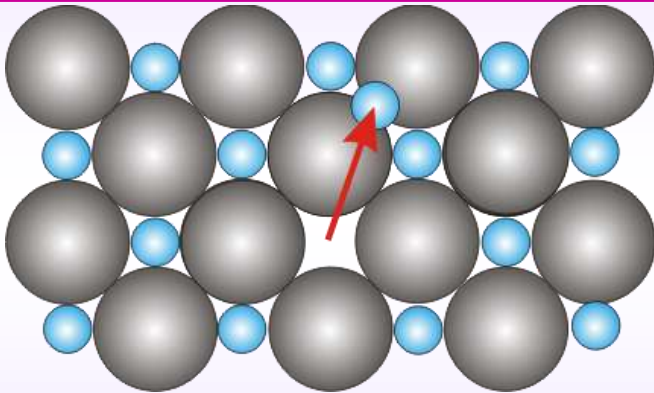
- ❑ This question can be addressed from the process/origin and concept (or energy calculation) perspectives. We consider the later now.
- ❑ Conceptually, we can create a vacancy by moving an atom from an atomic site to infinity.
- ❑ In practice (say during heating a substance), the atom will move from a atomic site to a ledge or a kink (*or other preferred sites*) on the surface. So the energy of formation is the difference in energy between these two states.
- ❑ Due to the creation of the vacancy the atoms around it will relax to a new configuration and correspondingly, the presence of the additional atom on the surface will cause changes locally (on the surface). These change will lead to a local change in electronic structure as well.
- ❑ Hence, the energy of the two states have to be computed taking into account these aspects.

Ionic Crystals

- ❑ In ionic crystals, during the formation of the defect the overall electrical neutrality has to be maintained (or to be more precise the cost of not maintaining electrical neutrality is high).

Frenkel defect

- Cation being smaller can get displaced to interstitial voids.
- This kind of self interstitial costs high energy in simple metals and is not usually found [$\Delta H_f(\text{vacancy}) \sim 1\text{eV}$; $\Delta H_f(\text{interstitial}) \sim 3\text{eV}$].
- E.g. in AgI & CaF₂ the cation can form a self interstitial.
 - Ag interstitial concentration near melting point:
 - in AgCl of 10^{-3} , ➤ in AgBr of 10^{-2} .

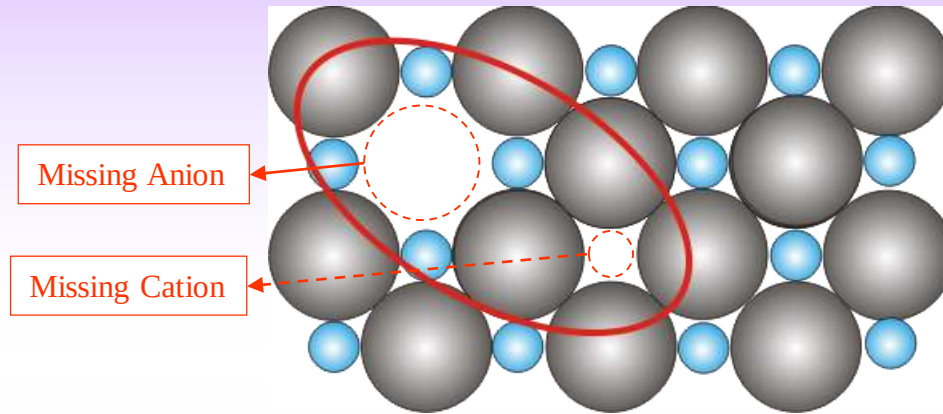


$$n_F \approx (NN_i)^{1/2} \exp\left(-\frac{\Delta H_s}{2kT}\right)$$

- $n_F \rightarrow$ no. of Frenkel defects in a MX crystal
- $\Delta H_F \rightarrow$ enthalpy of formation of a Frenkel defects
- $N_i \rightarrow$ no. of interstitial sites available

Schottky defect

- A Schottky defect consists of a pair of anion and cation vacancies → this maintains charge neutrality. This for example is found in Alkali halides.



- The total number of configurations is now the (number of ways the cation vacancy can be arranged) $\times$ (the number of ways the anion vacancy can be arranged). $\omega_{\text{total}} = \omega_{\text{anion}} \cdot \omega_{\text{cation}}$

$$\omega = \omega_{\text{anion}} \cdot \omega_{\text{cation}} = \left[\frac{N!}{n_v!(N-n_v)!} \right] \cdot \left[\frac{N!}{n_v!(N-n_v)!} \right] = \left[\frac{N!}{n_v!(N-n_v)!} \right]^2$$

$$\Delta S_{\text{config}} = k \ln \omega = 2k \ln \left[\frac{N!}{n_v!(N-n_v)!} \right]$$

$$\frac{n_s}{N} = \exp \left(- \frac{\Delta H_s}{2kT} \right)$$

Factor '2' $\Rightarrow$ the increase is steeper with 'T' as compared to vacancies

- $n_s \rightarrow$ no. of Schottky defects
- $\Delta H_s \rightarrow$ enthalpy of formation of a Schottky defect

❑ Typical enthalpies of formation of Schottky and Frenkel defects.

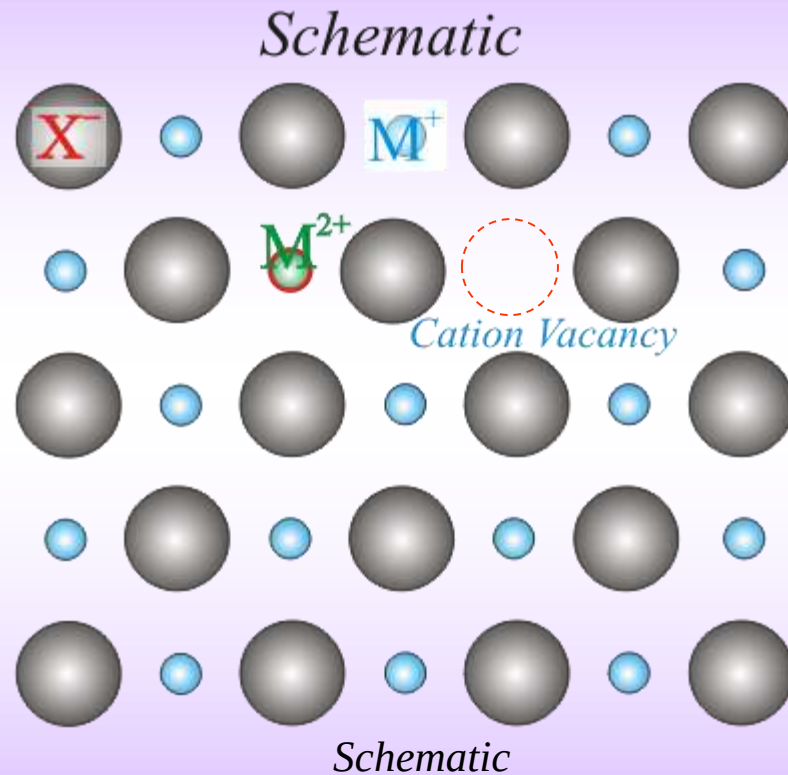
Schottky Defects		
Compound	ΔH_s (J) $\times 10^{-21}$	ΔH_s (eV)
LiI	2.08	1.30
LiBr	2.88	1.80
NaCl	3.69	2.30
CaO	9.77	6.10
MgO	10.57	6.60

Frenkel Defects		
Compound	ΔH_F (J) $\times 10^{-21}$	ΔH_s (eV)
β -AgI	1.12	0.70
AgBr	1.92	1.20
AgCl	2.56	1.60
CaF ₂	4.49	2.80
ZrO ₂	6.57	4.10

- ❑ Even in solids like LiI with low ΔH_s (enthalpy of formation of Schottky defects) (1.30 eV) the fraction of defects at RT (300K) is small (1.2×10^{-11}). At 1000K the fraction is 5.3×10^{-4} .
- ❑ This implies there are very few Schottky defects at RT.
- ❑ Depending on the values of ΔH_s & ΔH_F both these defects may be present in a crystal (though one of them dominates in most systems).

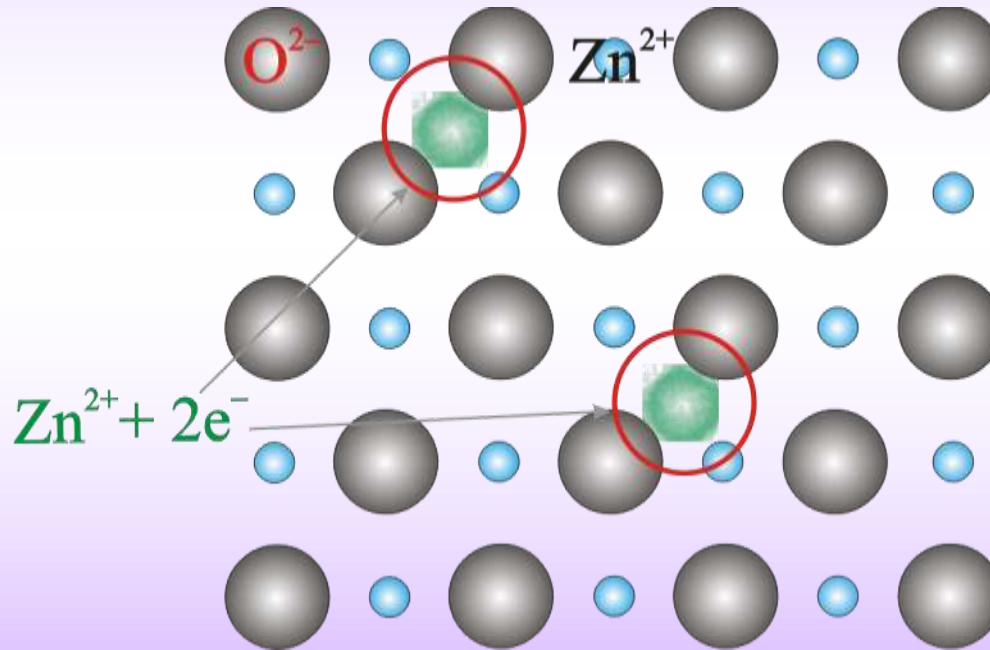
Other defects due to charge balance (/neutrality condition)

- If Cd^{2+} replaces Na^+ $\rightarrow$ one cation vacancy is created



Defects due to off stoichiometry

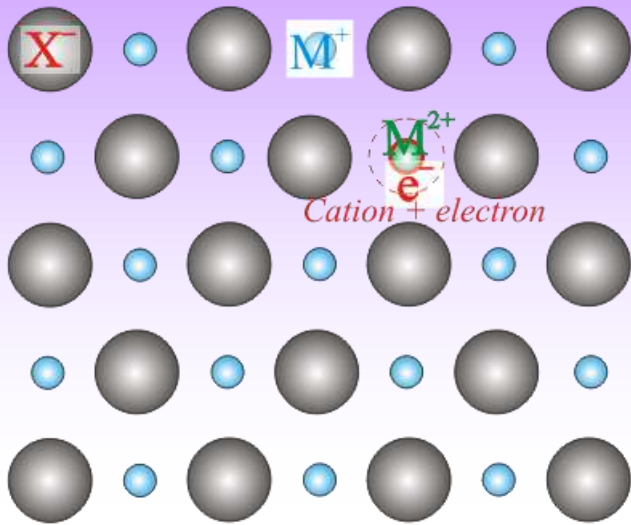
- ZnO heated in Zn vapour $\rightarrow \text{Zn}_y\text{O}$ ($y > 1$).
- The excess cations occupy interstitial voids.
- The electrons ($2e^-$) released stay associated to the interstitial cation.



Schematic

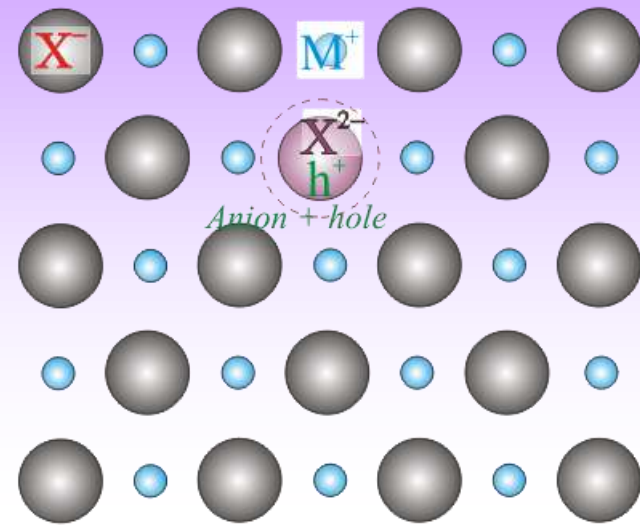
Other defect configurations: association of ions with electrons and holes

Schematic



M^{2+} cation associated with an electron

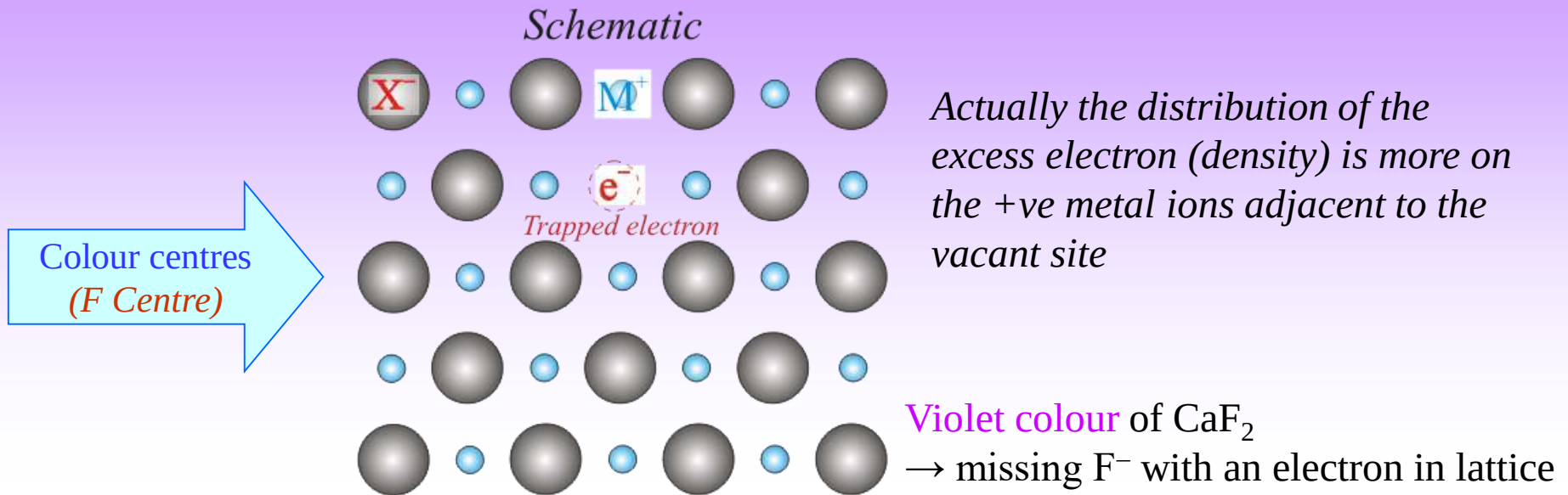
Schematic



X^{2-} anion associated with a hole

How do colours in some crystals arise due to colour centres (F-centres)?

F: Farbzentrum



Ionic Crystal	F centre absorption energy (eV)
LiCl	3.1
NaCl	2.7
KCl	2.2
CsCl	2.0
KBr	2.0
LiF	5.0

$$E = h\nu \quad E = h \frac{c}{\lambda}$$

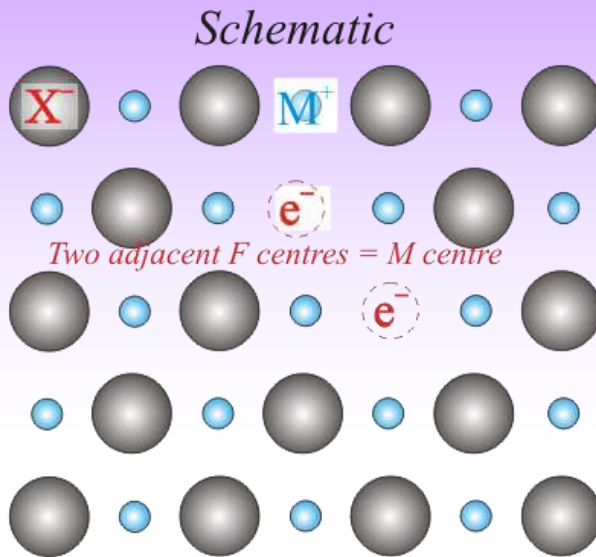
$$E_{\text{KBr}} = 2 \text{ eV} = 2 \times (1.602 \times 10^{-19}) \text{ J}$$

$$\lambda_{\text{KBr}}^{\text{absorption}} = \frac{(6.628 \times 10^{-34}) (3 \times 10^8)}{2 \times (1.602 \times 10^{-19})} = 6.2 \times 10^{-7} \text{ m} = 620 \text{ nm}$$

↑
Red

Visible spectrum: 390-750 nm

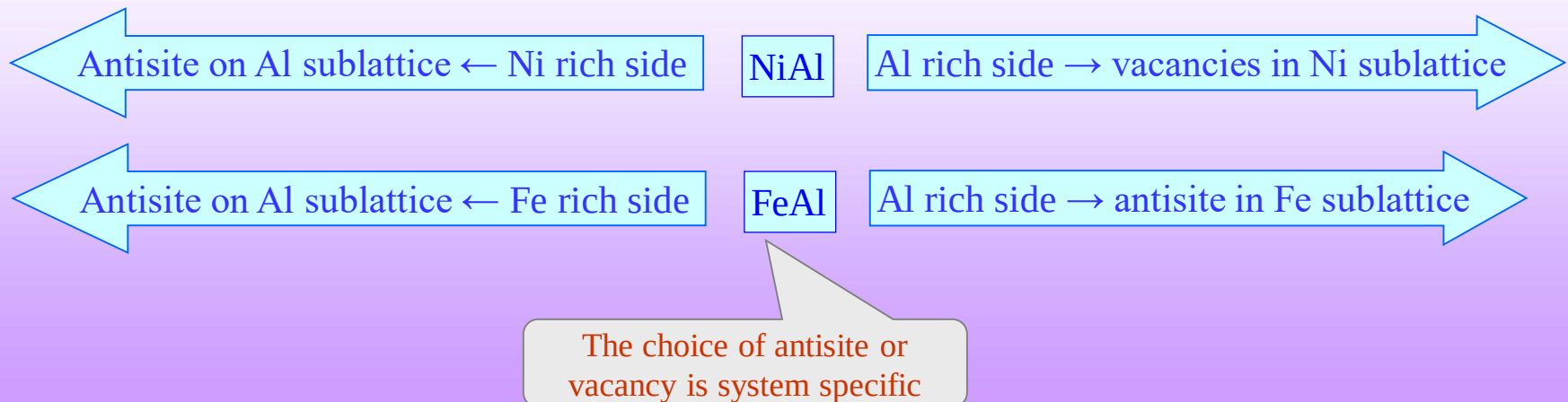
Some more complications: an example of defect association



Two adjacent **F centres** giving rise to a **M centre**

Structural Point defects

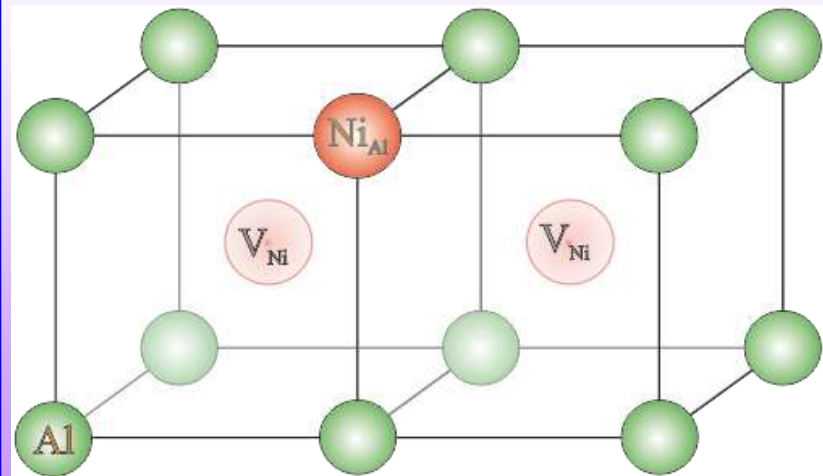
- ❑ In ordered **NiAl** (with ordered **B2** structure) → Al rich compositions result from vacancies in Ni sublattice.
- ❑ In Ferrous Oxide (**FeO**) with NaCl structure there is a large concentration of cation vacancies.
 - Some of the Fe is present in the Fe^{3+} state → correspondingly some of the positions in the Fe sublattice is vacant → leads to off stoichiometry (Fe_xO where x can be as low as 0.9 leading to considerable concentration of ‘non-equilibrium’ vacancies).
- ❑ In NaCl with small amount of Ca^{2+} impurity:
 - for each impurity ion there is a vacancy in the Na^+ sublattice.



Vacancies in ordered structures

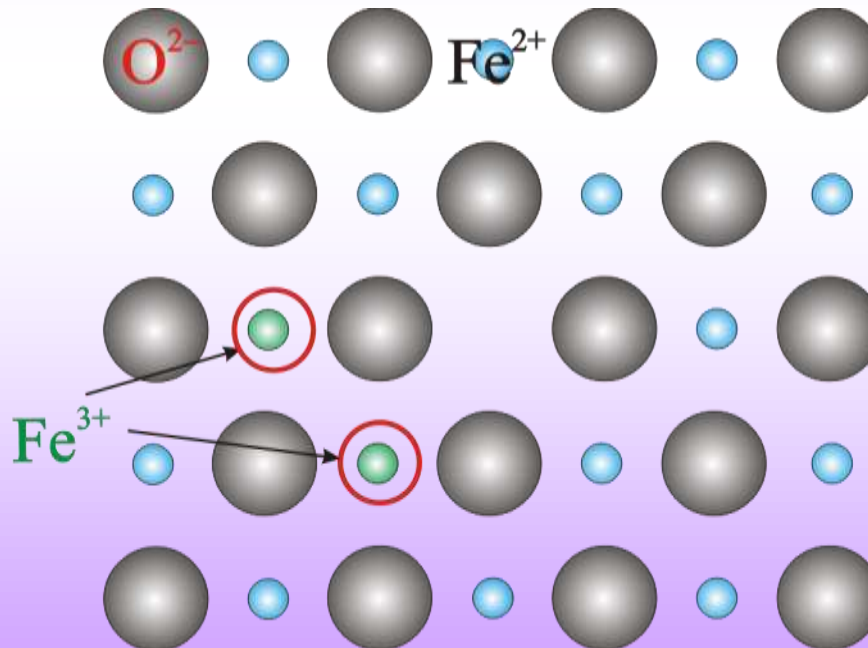
- ❑ With the example of NiAl, let us get a glimpse of some interesting possibilities which arise in ordered structures.
- ❑ We have already seen structural/constitutional defects in NiAl (antisite and vacancy). Now let us consider thermal (equilibrium) vacancies in NiAl.

- The pathway to understating the formation of a triple defect is as follows
 - Remove on Al atom from Al sublattice (Al_{Al}) and one Ni atom from Ni sublattice (Ni_{Ni}). This leaves the composition unchanged.
 - Now move a Ni atom to the vacant Al site (sublattice) to create an antisite defect (Ni_{Al}). This also does not change the stoichiometry (remains $\text{Ni}_{50}\text{Al}_{50}$).
 - In the end we 'land up' with two vacancies on Ni sublattice ($2V_{\text{Ni}}$) and one antisite defect (Ni_{Al}).
 - We have already seen that Al excess compositions are accommodated by vacancies (V_{Ni}) and Ni rich compositions by antisite defects (Ni_{Al}). *Both these are present in the scenario above.*



- FeO heated in oxygen atmosphere $\rightarrow \text{Fe}_x\text{O}$ ($x < 1$)
- Vacant cation sites are present
- Charge is compensated by conversion of ferrous to ferric ion:

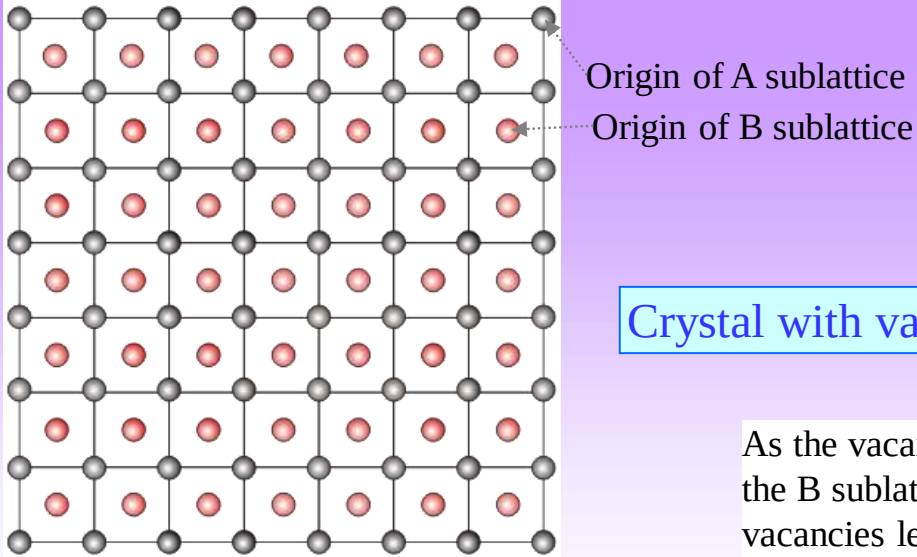
$$\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$$
- For every vacancy (of Fe cation) two ferrous ions are converted to ferric ions $\rightarrow$ provides the 2 electrons required by excess oxygen



Point Defect ordering

- ❑ Using the example of vacancies we illustrate the concept of defect ordering.
- ❑ As shown before, based on position vacancies can be random or ordered.
- ❑ Ordered vacancies (like other ordered defects) play a different role in the behaviour of the material as compared to random vacancies.

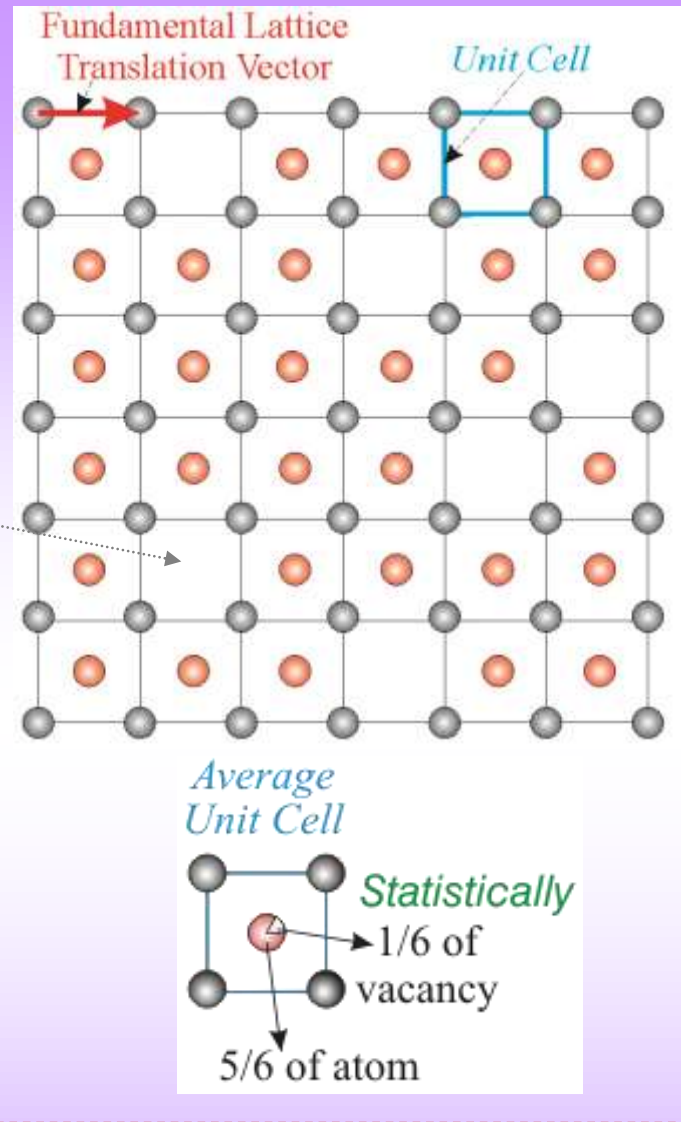
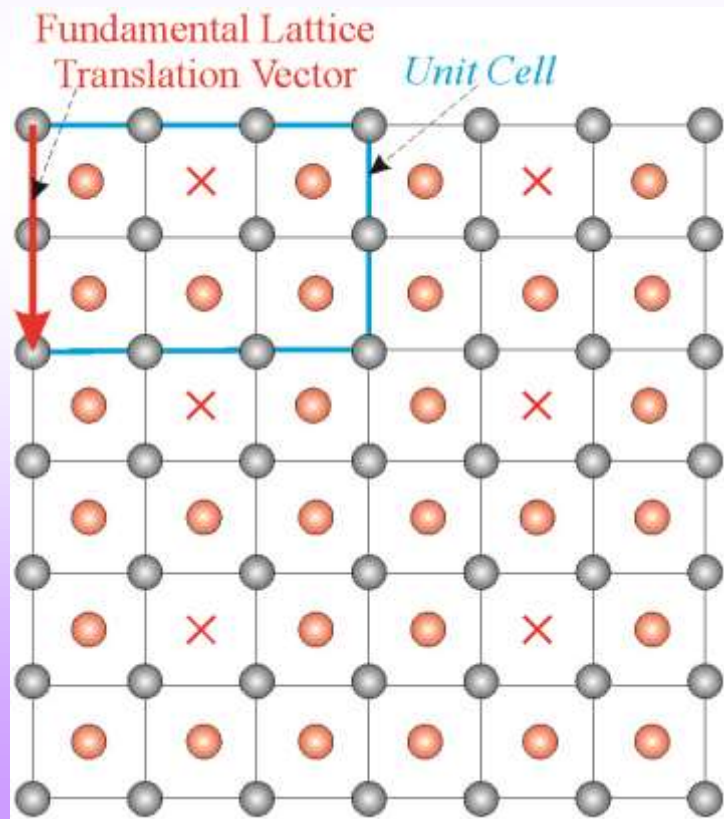
Schematic



Crystal with vacancies

As the vacancies are in the B sublattice these vacancies lead to off stoichiometry and hence are **structural vacancies**

Vacancy ordering



Examples of Vacancy Ordered Phases: V_6C_5 , V_8C_7

Vacancy Ordered Phases (VOP)

- ❑ Me_6C_5 trigonal ordered structures
(e.g. $\text{V}_6\text{C}_5 \rightarrow$ ordered trigonal structure exists between $\sim 1400\text{-}1520\text{K}$)
(The disordered structure is of NaCl type (FCC lattice) with C in non-metallic sites)
 - Space group: $P3_1$
 - The disordered FCC basis vectors are related to the ordered structure by:

$$a_{\text{trigonal}} = \frac{1}{2} \langle 21\bar{1} \rangle_{\text{FCC}}$$

$$b_{\text{trigonal}} = \frac{1}{2} \langle \bar{1}12 \rangle_{\text{FCC}}$$

$$c_{\text{trigonal}} = 2 \langle 1\bar{1}1 \rangle_{\text{FCC}}$$

Atom	Wyckoff Position	x	y	z
Vacancy	3(a)	1/9	8/9	1/6
C1	3(a)	4/9	5/9	1/6
C2	3(a)	7/9	2/9	1/6
C3	3(a)	1/9	5/9	1/3
C4	3(a)	4/9	2/9	1/3
C5	3(a)	7/9	8/9	1/3
V1	3(a)	1/9	5/9	1/12
V2	3(a)	4/9	2/9	1/12
V3	3(a)	7/9	8/9	1/12
V4	3(a)	1/9	2/9	1/6
V5	3(a)	4/9	8/9	1/6
V6	3(A)	7/9	5/9	1/6

Complex and Associated Point Defects

Association of Point defects (especially vacancies)

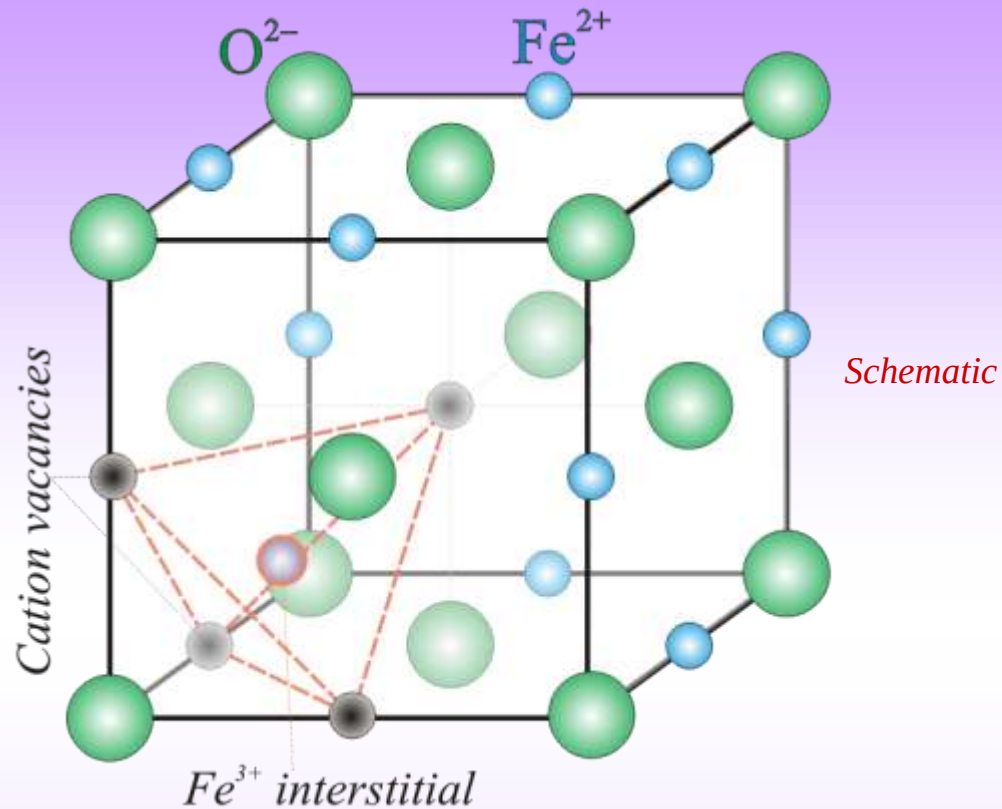
- ❑ Point defects can occur in isolation or could get associated with each other (we have already seen some examples of these).
- ❑ If the system is in equilibrium then the enthalpic and entropic effects (i.e. on G) have to be considered in understanding the association of vacancies.
- ❑
 - If two vacancies get associated with each other (forming a **di-vacancy**) then this can be visualized as a reduction in the number of bonds broken, leading to an energy benefit (in Au this binding energy is $\sim 0.3 \text{ eV}$).
 - but this reduces the number of configurations possible with only dissociated vacancies.
 - The ratio of mono-vacancies to divacancies increases with increasing temperature.
- ❑ Similarly an interstitial atom and a vacancy can come together to reduce the energy of the crystal $\rightarrow$ would preferred to be associated.
- ❑ Non-equilibrium concentration of interstitials and vacancies can condense into larger clusters.
 - In some cases these can be visualized as **prismatic dislocation loop** or **stacking fault tetrahedron**).
- ❑ Point defects can also be associated with other defects like dislocations, grain boundaries etc.
- ❑ We had considered a divacancy. Similar considerations come into play for **tri-vacancy** formation etc.

Complex Point Defect Structures: an example

- ❑ The defect structures especially ionic solids can be much more complicated than the simple picture presented before. Using an example such a possibility is shown.
- ❑ In transition metal oxides the composition is variable
- ❑ In NiO and CoO fractional deviations from stoichiometry (10^{-3} - 10^{-2})
→ accommodated by introduction of cation vacancies
- ❑ In FeO larger deviations from stoichiometry is observed
- ❑ At $T > 570^{\circ}\text{C}$ the stable composition is $\text{Fe}_{(1-x)}\text{O}$ [$x \in (0.05, 0.16)$]
- ❑ Such a deviation can 'in principle' be accommodated by Fe^{2+} vacancies or O^{2-} interstitials
- ❑ In reality the situation is more complicated and the iron deficient structure is the **4:1 cluster** → 4 Fe^{2+} vacancies as a tetrahedron + Fe^{3+} interstitial at centre of the tetrahedron + additional neighbouring Fe^{3+} interstitials
- ❑ These 4:1 clusters can further associate to form 6:2 and 13:4 aggregates

Note: these are structural vacancies

Continued...



4:1 cluster $\rightarrow$ 4 Fe²⁺ vacancies as a tetrahedron + Fe³⁺ interstitial at centre of the tetrahedron + additional neighbouring Fe³⁺ interstitials

The figure shows an ideal starting configuration- the actual structure will be distorted with respect to this depiction

Methods of producing point defects

❑ Growth and synthesis

Impurities may be added to the material during synthesis.

❑ Thermal & thermochemical treatments and other stimuli

- Heating to high temperature and quench
- Heating in reactive atmosphere
- Heating in vacuum → e.g. in oxides it may lead to loss of oxygen
- Etc.

❑ Plastic Deformation

❑ Ion implantation and irradiation

- Electron irradiation (typically $>1\text{MeV}$)
 - Direct momentum transfer or during relaxation of electronic excitations)
- Ion beam implantation (As, B etc.)
- Neutron irradiation.

Solved Example

What is the equilibrium concentration of vacancies at 800K in Cu

Data for Cu:

- Melting point = 1083 °C = 1356K
- ΔH_f (Cu vacancy) = 120 × 10³ J/mole
- k (Boltzmann constant) = 1.38 × 10⁻²³ J/K
- R (Gas constant) = 8.314 J/mole/K

*First point we note is that we are below the melting point of Cu
800K ~ 0.59 T_m(Cu)*

$$\frac{n}{N} = \exp\left(-\frac{\Delta H_f}{RT}\right)$$

$$\frac{n}{N} = \exp\left(-\frac{120 \times 10^3}{8.314 \times 800}\right) = \exp(-18.04) = 1.46 \times 10^{-8}$$

If we increase the temperature to 1350K (near MP of copper)

$$\frac{n}{N} = \exp\left(-\frac{120 \times 10^3}{8.314 \times 1350}\right) = \exp(-10.69) = 2.27 \times 10^{-5}$$

Experimental value: 1.0 × 10⁻⁴

Solved Example

If a copper rod is heated from 0K to 1250K increases in length by ~2%. What fraction of this increase in length is due to the formation of vacancies?

Data for Cu:

- ΔH_f (Cu vacancy) = 120×10^3 J/mole
- R (Gas constant) = 8.314 J/mole/K

Cu is FCC ($n = 4$)

$$\text{Fractional increase in length} = \left(\frac{L - L_0}{L_0} \right) = 0.02 \quad \Rightarrow \Delta L = 0.02 L_0, \text{ where subscript 0 refers}$$

to the 0K state.

There are two contributions to this increase in length (ΔL): (i) from thermal expansion (ΔL_{TE}) and (ii) from increase in fraction of vacancies due to heating (ΔL_v). The vacancies are created by atoms migrating to the surface leading to an increase in volume of the material. The vacancies are incorporated in the crystal due to the entropic stabilization that it provides (which more than offsets the increase in enthalpy caused by broken bonds).

$$V = L^3 \Rightarrow dV = 3L^2 dL$$

$$\Rightarrow \frac{dV}{V} = 3 \frac{dL}{L} \quad \text{or in terms of finite differences: } \frac{\Delta V}{V} = 3 \frac{\Delta L}{L} \quad (1)$$

The fraction required to be calculated is $\rightarrow f = \frac{\Delta L_v}{\Delta L}$

$$\frac{n_v}{N_0} = \exp \left[\frac{-120 \times 10^3}{8.314 \times 1250} \right] = \exp(-11.54) = 9.7 \times 10^{-6} (= x)$$

Continued...

1 unit cell gives a volume $\uparrow$ of $a^3 \Rightarrow 4$ vacancies give a volume of a^3

$\Rightarrow n_v$ vacancies give a volume of $\frac{n_v a^3}{4} = \Delta V_v$

Equation (1) $\Rightarrow \frac{\Delta V_v}{V_0} = \frac{3\Delta L_v}{L_0}$. Where V_0 is given by: $V_0 = \frac{N_0 a^3}{4}$

$$\Rightarrow \frac{\Delta V_v}{V_0} = \frac{\left(\frac{n_v a^3}{4}\right)}{\left(\frac{N_0 a^3}{4}\right)} = \frac{\left(\frac{x N_0 a^3}{4}\right)}{\left(\frac{N_0 a^3}{4}\right)} = x \Rightarrow \frac{3\Delta L_v}{L_0} = x, \left(\frac{\Delta L_v}{L_0} = \frac{x}{3}\right)$$

$$f = \frac{\left(\frac{x}{3} L_0\right)}{\Delta L} = \frac{\left(\frac{x L_0}{3}\right)}{0.02 L_0} = \frac{9.7 \times 10^{-6}}{3(0.02)} = 1.6 \times 10^{-4}$$

$\Rightarrow$ this effect is about 1 in 10^4 as compared to thermal expansion due to atomic vibrations!

- ❑ These 3 words are technical terms in materials science and are often used in more than one context.
- ❑ Vacancy is typically a missing atom from its site, but is sometimes used in the context of a missing electron from its shell (*“vacancy in the L shell”*).
- ❑ Void can come in two forms: (a) inter-atomic voids in crystals (*the crystallographic voids*) and (b) ‘macroscopic’* void (which is missing matter in a material).
- ❑ A hole is a missing electron in the valence band. Instead of treating the $(n-1)$ negatively electrons in the valence band, we consider a positively charged hole (in the valence band).

* Macroscopic as compared to the inter-atomic void.